What Is the "1.8/2.0 (EM)" Chiral Interaction?

A Novice-Friendly Explanation

Based on Sun et al., *Phys. Rev. X* **15**, 011028 (2025)
and Hebeler et al., *Phys. Rev. C* **83**, 031301 (2011)

The Short Answer

The "1.8/2.0 (EM)" is a specific recipe for the force between protons and neutrons inside a nucleus. It was assembled by Hebeler, Bogner, Furnstahl, Nogga, and Schwenk in 2011, and it has become one of the most successful and widely used nuclear interactions in modern *ab initio* (first-principles) nuclear physics.

The name encodes three things:

1.8	=	momentum cutoff for the two-nucleon force (in fm^{-1})
2.0	=	momentum cutoff for the three-nucleon force (in fm^{-1})
(EM)	=	Entem & Machleidt, who built the original two-nucleon potential

Let us unpack each piece.

Step 1: Where Do Nuclear Forces Come From?

The ultimate theory of the strong force is **quantum chromodynamics (QCD)**—the theory of quarks and gluons. But QCD is impossibly hard to solve at the energies relevant to nuclear physics. So physicists use an approximation strategy called **chiral effective field theory (chiral EFT)**.

> Analogy: A City Map Imagine you want to drive across a city. You do not need a map showing every brick in every building—you need a *road map* that captures the features relevant at your scale. Chiral EFT is like that road map: it describes nuclear forces in terms of the particles that matter at nuclear energy scales (protons, neutrons, and pions) rather than the underlying quarks and gluons.

Chiral EFT organises the nuclear force as a series of increasingly detailed corrections, like zooming in:

Order	Label	What it adds
Leading	LO	One-pion exchange + simple contacts
Next-to-leading	NLO	Two-pion exchange starts
Next-to-next-to-leading	NNLO	More two-pion exchange + 3-nucleon forces appear
$\mathrm{N^3LO}$	$\mathrm{N^3LO}$	Refined two-pion exchange + more contacts

The two-nucleon (NN) potential used in the 1.8/2.0 (EM) interaction was derived at **$\mathrm{N^3LO}$** by Entem and Machleidt (hence "EM")—a very high-accuracy description of how

two nucleons interact.

Step 2: Why "Soften" the Force? (The 1.8)

The raw N^{3}LO potential from Entem & Machleidt is very "hard"—it contains strong, short-distance (high-momentum) components that make many-body calculations extremely difficult and slow to converge.

To fix this, the potential is processed through a mathematical tool called the **Similarity Renormalization Group (SRG)**. The SRG smoothly dials down the high-momentum parts of the interaction without changing the physics at the energies you care about.

Analogy: Noise Reduction on a Photo Imagine a very detailed but noisy photograph. You apply a filter that removes the high-frequency noise (the grainy pixels) while keeping the meaningful features (the shapes, colours, faces) intact. That is essentially what SRG does to the nuclear force. The "resolution" of the smoothed potential is set by a momentum cutoff.

The "1.8" in Detail The two-nucleon force is SRG-evolved down to a momentum cutoff of $\mathbf{1.8\,fm^{-1}}$.
In physical terms: interactions involving relative momenta above $\sim 1.8\,\mathrm{fm}^{-1}$ (i.e., very short-distance, hard collisions) are "integrated out"—their effects are folded into the remaining lower-momentum interaction. The result is a much softer potential that converges rapidly in many-body calculations.

Step 3: The Three-Nucleon Force (The 2.0)

When three nucleons are close together, they experience a force that cannot be described as just the sum of pairwise (two-body) forces. This genuine **three-nucleon force (3NF)** is a prediction of chiral EFT—it arises naturally at NNLO and is essential for reproducing nuclear binding energies and the location of the nuclear drip lines.

Analogy: Three People on a Trampoline If two people stand on a trampoline, they each feel the dip the other creates. But if a third person joins, the surface deforms in a way that is *not* just the sum of the two-person dips—there is a genuinely new three-body effect from how the surface collectively distorts. Three-nucleon forces are the nuclear equivalent.

The "2.0" in Detail The three-nucleon force is included at **NNLO** with a nonlocal regulator and a momentum cutoff of $\mathbf{2.0\,fm^{-1}}$.
Unlike the two-body force, this part is *not* SRG-softened—it is used in its original ("bare") form. The cutoff $2.0\,\mathrm{fm}^{-1}$ simply defines the resolution scale at which the three-nucleon force is regulated.

Putting It All Together

1.8/2.0 (EM) =

NN force from Entem & Machleidt at N^{3}LO, SRG-softened to 1.8 fm^{-1}

+

3N force at NNLO, bare, with cutoff 2.0 fm^{-1}

Why Is This Interaction So Popular?

Since its introduction in 2011, the 1.8/2.0 (EM) interaction has been used in hundreds of *ab initio* nuclear structure calculations. Its popularity comes from a remarkable track record:

- It reproduces **binding energies** across a huge range of nuclei—from helium ($A = 4$) through lead ($A = 208$).
- It gives accurate **energy spectra** (the pattern of excited states) in light, medium, and heavy nuclei.
- It predicts **charge radii** and other observables that agree well with experiment.
- The SRG softening makes calculations **computationally feasible** even for heavy systems.

In the Sun et al. paper, it is used for "point predictions"—their single best-estimate calculations of rotational bands and $B(E2)$ transitions in neon isotopes.

Are There Any Caveats?

Yes—and the paper is careful about this:

1. **SRG changes collective properties.** The softening process that makes the NN force manageable can modify things like the nuclear moment of inertia, which directly affects rotational spectra. This is why the authors *also* use a separate ensemble of un-softened interactions (the ΔNNLO set) for uncertainty quantification.
2. **Induced many-body forces.** In principle, the SRG transformation generates new three-body and higher-body forces. The 1.8/2.0 (EM) interaction does not fully account for these induced terms, which is a known source of systematic uncertainty.
3. **Not a unique choice.** Other interactions exist with different cutoffs, different chiral orders, or different fitting strategies. No single interaction is "the right one"—each represents a particular approximation, and comparing results across multiple interactions is how the field estimates theoretical uncertainty.

Quick Reference Card

Property	**Value / Description**
Full name	1.8/2.0 (EM)
Original NN potential	Entem & Machleidt, N^3LO (2003)
NN softening method	Similarity Renormalization Group (SRG)
NN momentum cutoff	$1.8\,\mathrm{fm}^{-1}$
3N force order	NNLO
3N regulator	Nonlocal, cutoff = $2.0\,\mathrm{fm}^{-1}$
3N softening	None (bare)
Assembled by	Hebeler, Bogner, Furnstahl, Nogga, Schwenk (2011)
Reference	*Phys. Rev. C* **83**, 031301(R)

Bottom Line The 1.8/2.0 (EM) interaction is a carefully constructed nuclear force that combines a high-precision two-nucleon potential (softened for computational tractability) with a genuine three-nucleon force from chiral effective field theory. It serves as the workhorse interaction for modern *ab initio* nuclear structure calculations because it balances accuracy against computational cost, and it has an outstanding track record of reproducing experimental nuclear properties across the periodic table.